Exam Calculus A

202001171 (5 EC) / 202001172 (4 EC)

10 March 2025
18:15-21:15h

This exam consists of 8 exercises for 202001172 and of 10 exercises for 202001171.
Motivate all your answers.

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1. Where is the function $f(x)$ continuous?

$$f(x) = \begin{cases} \frac{x^3-8}{x^2-4}, & x \neq 2, x \neq -2 \\ 3, & x = 2 \\ 4, & x = -2 \end{cases}$$

2. Find the derivative of

$$x\cos(2x+3y) = y\sin x$$

3. Find the extreme values (absolute and local) of the function $g(x) = x^2\sqrt{5-x}$. Also write down where the extreme values occur, and specify per extreme value whether it is an absolute or local minimum or maximum.

4. Evaluate the following integrals

   (a)
$$\int_{-1}^{-1/2} t^{-2}\sin^2\left(1+\frac{1}{t}\right)\,dt$$

   (b)
$$\int \sin 2x\cos 4x\,dx$$

   (c)
$$\int_0^1 \left[\frac{2}{\sqrt{1-t^2}}\mathbf{i} + \frac{\sqrt{3}}{1+t^2}\mathbf{k}\right]\,dt$$

5. Let $\mathbf{v} = -\mathbf{i}+\mathbf{j}$ and $\mathbf{u} = \sqrt{2}\mathbf{i}+\sqrt{3}\mathbf{j}+2\mathbf{k}$. Find

   (a) $\mathbf{v}\cdot\mathbf{u}$
   (b) the vector $\text{proj}_{\mathbf{v}}\mathbf{u}$
   (c) $\mathbf{v}\times\mathbf{u}$

6. Let $x = 4\sin t,\ y = 5\cos t,\ 0 \leq t \leq 2\pi$ describe the motion of a particle in the xy-plane. Identify the particle's path by finding a Cartesian equation for it. Graph the Cartesian equation. Indicate the portion of the graph traced by the particle and the direction of motion.

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7. Find parametric equations for the line tangent to the curve of intersection of the surfaces $xyz = 1,\ x^2 + 2y^2 + 3z^2 = 6$ at the point $(1, 1, 1)$.

8. Solve the following initial value problem

$$(x+1)\frac{dy}{dx} - 2(x^2+x)y = \frac{e^{x^2}}{x+1},\ x > -1,\ y(0) = 5$$

Exercises 9 and 10 are only for 202001171.

9. Find all solutions of the equation $x^4 + 9x^2 + 81 = 0$. Give your answer in Cartesian form.

10. Solve the following differential equation

$$y'' + 2y' = x^2 - e^x$$

Norm:

Exercise	1	2	3	4a	4b	4c	5	6	7	8	9	10	total
202001171	3	4	4	3	4	3	4	4	3	4	5	4	45
202001172	3	4	4	3	4	3	4	4	3	4	-	-	36

Exam grade 202001171: 1 + (obtained points)/5
Exam grade 202001172: 1 + (obtained points)/4